6. CONVOLUTION AND SAMPLING TECHNIQUES

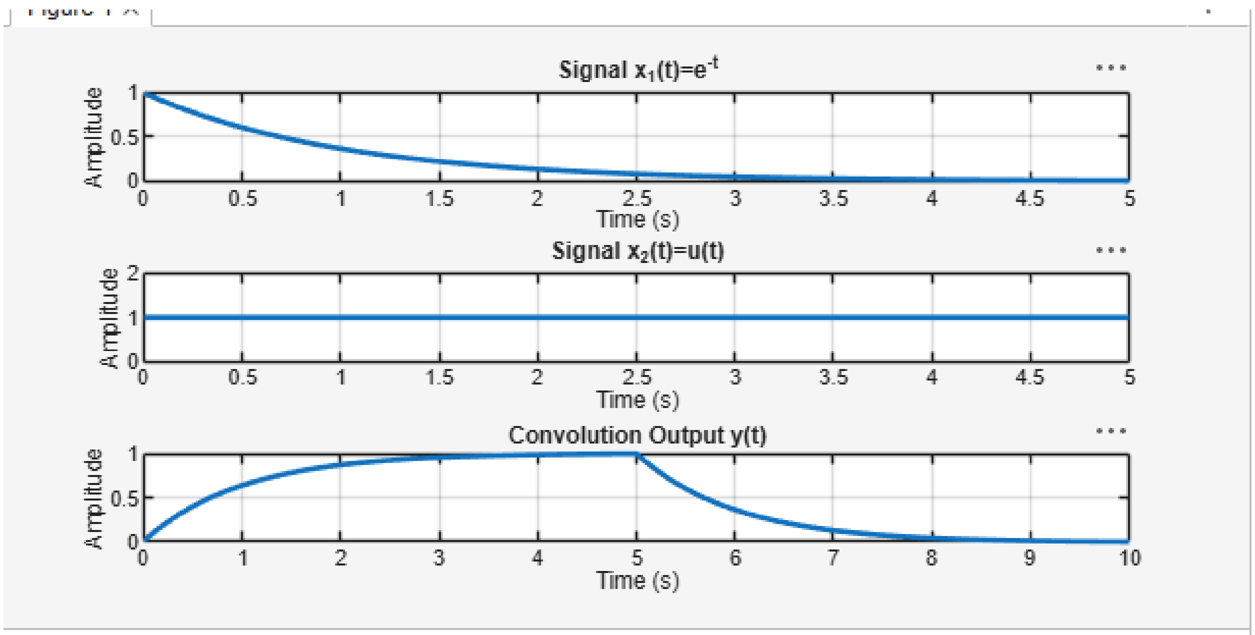
Experiment 6.1

Convolution of Continuous Time Signals

Proogram:

```
% Convolution of Continuous-Time Signals
clc;
clear;
close all;
% Time axis
t = 0:0.01:5;
% Input signals
x1 = exp(-t); % Exponential signal
x2 = ones(size(t)); % Unit step signal
% Convolution
y = conv(x1,x2)*0.01;
% Time axis for output
t_conv = 0:0.01:(length(y)-1)*0.01;
% Plotting
figure;

subplot(3,1,1);
plot(t,x1,'LineWidth',2);
grid on;
title('Signal x_1(t)=e^{-t}');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,2);
plot(t,x2,'LineWidth',2);
grid on;
title('Signal x_2(t)=u(t)');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,3);
plot(t_conv,y,'LineWidth',2);
grid on;
title('Convolution Output y(t)');
xlabel('Time (s)');
ylabel('Amplitude');
```



Experiment 6.2

Convolution of Discrete Time Signals

Program:

```
% Convolution of Discrete-Time Signals
clc;
clear;
close all;
% Input sequences
x1 = [1 2 3 4];
x2 = [1 1 1];
% Convolution
y = conv(x1,x2);
% Sample indices
n1 = 0:length(x1)-1;
n2 = 0:length(x2)-1;
ny = 0:length(y)-1;
% Display output
disp('Convolution Output:');
disp(y);
% Plotting
figure;
subplot(3,1,1);
stem(n1,x1,'filled');
grid on;
title('Input Sequence x_1(n)');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,2);
stem(n2,x2,'filled');
```

```
grid on;
title('Input Sequence x_2(n)');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,3);
stem(ny,y,'filled');
grid on;
title('Convolution Output y(n)');
xlabel('n');
ylabel('Amplitude');
```